# The Mosaic of Human-AI Co-Creation: Emerging human-technology relationships in a co-design process with generative AI

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## Abstract

This study explored how pre-service teachers (N=33) perceived human-technology relationships with generative AI (genAI). The study employed a research-creation approach and implemented a hands-on workshop, in which the participants engaged in a speculative design process using generative AI. The study focused on how participants, armed with their new tool, approached their designs, made design decisions, and interacted with the responsive tool. The qualitative analysis of the video data from students' project presentations employed thematic analysis, interpreting the students' responses in relational terms. The results revealed that the emerging human-technology relationships were primarily expressed through distributed decision-making, with the AI actively contributing both to the object of activity and to the emerging design process. The findings highlight that genAI is neither passive nor neutral tools but actively transforms both the design process and its outcomes, shaping how people experience new forms of agency in relation to such technology.

**keywords:** artificial intelligence, text-to-image generative models, generative AI, prompt crafting, prompting, human-technology relationships, teacher education

## Introduction

In 2022, the emergence of easy-to-use technologies like Dall-E and Midjourney made text-to-image generative models increasingly popular for creating visual content, designs, and artwork (Dhariwal & Nichol, 2021; Liu & Chilton, 2022; Oppenlaender, 2022). While the use of machine learning (ML) in creative work traditionally required a deep knowledge of ML and programming skills (Fiebrink, 2019), the advent of these new tools has enabled users to generate impressive and highly aesthetic images using natural language (Oppenlaender, 2022; Vartiainen et al., 2023; Vartiainen & Tedre, 2023). Consequently, the widespread interest in and popularity of these tools have sparked passionate discussions and debates on how existing practices and the future

of creative fields are being reshaped by the use of generative AI (genAI) tools (Audry, 2021). These models are employed in the creation of digital art, text, music, and videos, as well as in the fusion of media in various creative endeavors. Those uses have raised new questions and ethical concerns related to copyright, privacy, and ownership, and have introduced many other complex questions encompassing societal, economic, environmental, legal, and ethical issues (Bommasani et al., 2021). Thus, it has become evident that these tools carry implications for educational institutions and practices (Vartiainen & Tedre, 2023).

Insofar as genAI is transforming human-technology relationships, a question arises: How can one ensure that people understand how genAI generates content and how data-driven tools change the nature of interactive processes, giving rise to new ethical, environmental, and societal questions? While some early research exists on how designers and artists use AI in their creative practices (Audry, 2021; Caramiaux & Fdili Alaoui, 2022; Fiebrink, 2019), research and theorization on the use of genAI in art, craft, design, and technology education is still in its infancy (Vartiainen et al., 2023; Vartiainen & Tedre, 2023). Researchers in the field of art and design have called for further research on the relationships between human and non-human actors in creative practice (e.g. Cypher, 2017). Moreover, little is known about the genAI experiences of teachers and pre-service teachers, who play a crucial role in educating future generations of designers, makers, and citizens of data-driven society.

Relying on sociocultural theories of mediated action and post-phenomenological approaches in the philosophy of technology, this article aims to address this gap by providing empirical findings and theoretical explorations of how pre-service craft teachers experience human-technology relationships with genAI. Following the research-creation approach (Lupton & Watson, 2022), we organized a hands-on workshop for pre-service teachers in Finland. The participants first engaged in a speculative design process using genAI, and then reflected on how they approached their designs in that technological setting, how they made design decisions, and how they interacted with the tool. The paper addresses the following research question: *How do pre-service teachers experience human-technology relationships in a co-design process with generative AI?*

## Theoretical framework

In the field of craft and design education, the interactions between humans and technological artifacts are often approached through socio-cultural and cultural-historical activity theories. In general, these theories posit that human actions and thoughts are profoundly shaped by the "cultural tools" or "mediational means" that individuals and groups employ within specific social activities (Vygotsky, 1978; Wertsch, 1993). This perspective sees that human interaction with the world occurs

indirectly and is also shaped by social interactions and the context of action (Wertsch, 2007).

Traditional socio-cultural theories have been criticized for their tendency to emphasize the active role of the human subjects and treat tools and technologies as passive backdrops in human actions (Fenwick et al., 2011; Verbeek, 2011). Meanwhile, human-technology relations have piqued the interest of many thinkers beyond the realm of education, in fields ranging from the philosophy of technology to postphenomenological approaches in science and technology studies (STS) (Verbeek, 2011). Don Ihde (1990) and his followers, for example, conceptualized technologies as mediators that shape people's perceptions, experiences, practices, and, ultimately, people's relationships with the world.

The insights above do not privilege human agency and intentionality over tools and technologies, but focus on exploring how human and non-human relationships can directly impact practice, knowing, and decision-making (Verbeek, 2011, Fenwick, 2011). For example, the advent of computers has fundamentally revolutionized the epistemic practices of science by providing computational power, new tools, and methods that help scientists to explore, create, evaluate and share knowledge about the world in new ways (Denning & Tedre, 2019). The same can be said of human behavior and interactions as technologies shape everyday decision-making for individuals in various ways (Verbeek, 2011), whether that be the music people listen to or products they buy online, the information they receive in search results, or impactful decisions they make when searching for an apartment or a dating partner. As Ihde (1990) noted, technology is often taken for granted in the realm of people's daily activities, and people often barely notice it, even though technology has become a lens through which they see the world.

The process of design and learning can also be viewed as emerging from a dynamic assemblage of human and non-human entities that interact with each other (Fenwick et al., 2011; Taguchi, 2011). This conceptualization shifts the attention to how different configurations are formed, how decision-making is distributed, and how these hybrid relationships influence emerging actions (Fenwick, 2011, Verbeek, 2011). In that context mediated action refers not only to the means that people utilize to act upon the object of their activities, but also to how technologies shape the ways people think, act, and collaborate when forming goals, making design decisions, and intentionally acting to achieve those goals. This approach to mediation goes beyond conceptualizing tools as passive instruments, and it acknowledges that, by altering one's understanding of the world and behavior, these technologies influence the core features of human agency (Verbeek, 2011). It further recognizes that actions of each actor within the interconnected network mutually influence each other: Users shape technology that, in turn, shapes users (Rahwan et al., 2019). Fenwick (2011) added that

human and nonhuman relations are not fixed properties, but actively configure, constrain, enable, and close off specific opportunities and forms of action.

With the rise of genAI posing new questions about human-technology relationships, those questions often stem from new epistemological and ontological tensions. Firstly, regarding epistemology, the rise of genAI highlights the boundary between propositional knowledge (know-that) and procedural knowledge (know-how), as well as between ability, knowledge, and understanding (Darwiche, 2018). GenAI tools can generate pictures by relying merely on a statistical model of a phenomenon, such as pixel patterns related to certain text labels, without understanding their meaning (Saharia et al., 2022). Major generative tasks can be conducted through just statistical pattern matching, with no qualitative understanding, intuition, lived experience, judgment, or sense of meaning (Smith, 2019). Secondly, regarding ontology, genAI systems exemplify the blurring of boundaries between agents and objects of action: They can perform tasks traditionally associated with cognition (Darwiche, 2018), wield decision-making authority, and further delegate decision-making to other non-human systems. Whether genAI represents a new category of agency or creativity is a debated topic, with viewpoints ranging from denying the category to AI altogether, to viewing AI-human teaming as a new type of collaborative creativity, and arguing for parallels between aspects of human and AI creativity (Anantrasirichai & Bull, 2022; Boden, 2009, 2010; Cetinic & She, 2022; Cypher, 2017; Hertzmann, 2020).

Insofar as human activities cannot be separated from the mediating tools and technologies that help people to construct ideas and knowledge about the world, a better understanding of the interconnected nature of human and nonhuman relations is paramount for agentive actions in the age of generative AI. To further that aim, the next sections present results from an intervention that explored the development of design ideas and the experiences of agentive actions in light of the human and non-human relationships that emerged when future teachers engaged in the process of collaborative design with generative AI.

# Materials and methods

## Setting and participants

The design of this study was based on our previous work and design-based research on integrating AI topics into primary and tertiary education (Vartiainen et al., 2021, 2022, 2023; Vartiainen & Tedre, 2023). The data were collected as part of a higher education course in Finland in the fall of 2023. The participants were 33 pre-service teachers who had chosen craft science as their major or minor subject as part of their 5-year MA program in Education.

The research complied with the recommendations of the Finnish National Board on Research Integrity (TENK, 2019). All data collection procedures received institutional approval from the administration of the Department of Teacher Education, and a participant information sheet was disseminated to course participants. The voluntary nature of participation and the right to withdraw were emphasized and discussed with the students when implementing the intervention.

The intervention started with a two-hour lecture, which introduced the students to AI in general and the mechanisms of text-to-image generative models in particular. It also provided insights into the societal impact and ethical concerns related to AI. After the lecture, students participated in a four-hour workshop. Three identical workshops were organized on three consecutive days, with each student participating in one. The workshop employed a research-creation approach to engage participants in the creative process of making digital artifacts. The approach was intended to inspire discussions and reflections on the complicated human-technology relationships involved (Lupton & Watson, 2022).

In the workshop, the students were organized into small groups of 3-4 students. After an introduction to biomimetic design (Benyus, 2002) and the speculative approach (Auger, 2013), they were presented with an imaginary future dystopian scenario depicting challengingly dark and wet weather. Groups were instructed to identify possible problems stemming from the scenario, and given an open-ended task to collaboratively design a speculative biomimetic product to tackle one such problem. For the ideation and design process, each group was provided with a laptop, briefed on text-to-image prompting, and tasked to explore and experiment with text-to-image genAI. Further scaffolding and support was provided based on each group's needs as the collaborative processes progressed. Following the ideation and design phase, the small groups presented their detailed product designs and were further encouraged to reflect on their collaborative design process and the emerging relationships with genAI.

## Data collection

Similar to the study by Lupton and Watson (2022), applying creative making in the context of a speculative design task aimed to prompt the participants to share their experiences and develop new ideas, rather than simply answering a set of interview questions. Data were collected from students' reflective discussions, presentations, and written course assigments. The main data set consisted of 161.5 minutes of video recordings of students' project presentations and joint reflections. Moreover, before and after the intervention, students were given an individual assignment to write down their thoughts on the role of tools and technologies in the design process. These written reflections served as supplementary data.

## Data analysis

Video data from the project presentations and joint reflections were transcribed, anonymized, and imported into the qualitative analysis software Atlas.ti. The data were analyzed using thematic analysis (Braun & Clarke, 2006) to understand how the participants experienced human-technology relationships. Initially, the data were read several times to gain a general sense of students' reflections on their design processes. Subsequently, meaningful units related to the research question were given preliminary, descriptive names. In order to identify how these units could form overarching, candidate themes, these data-driven interpretations were then subjected to a further analysis, in which different codes were compared and contrasted. The final phase involved reviewing potential themes in light of theories on human-technology relationships, which helped to conceptualize the emerging human and non-human relationships. To improve the reliability of the data analysis, these themes were negotiated between the researchers. Several examples from the data are provided to allow readers evaluate these interpretations (Hammer & Berland, 2014).

# Results

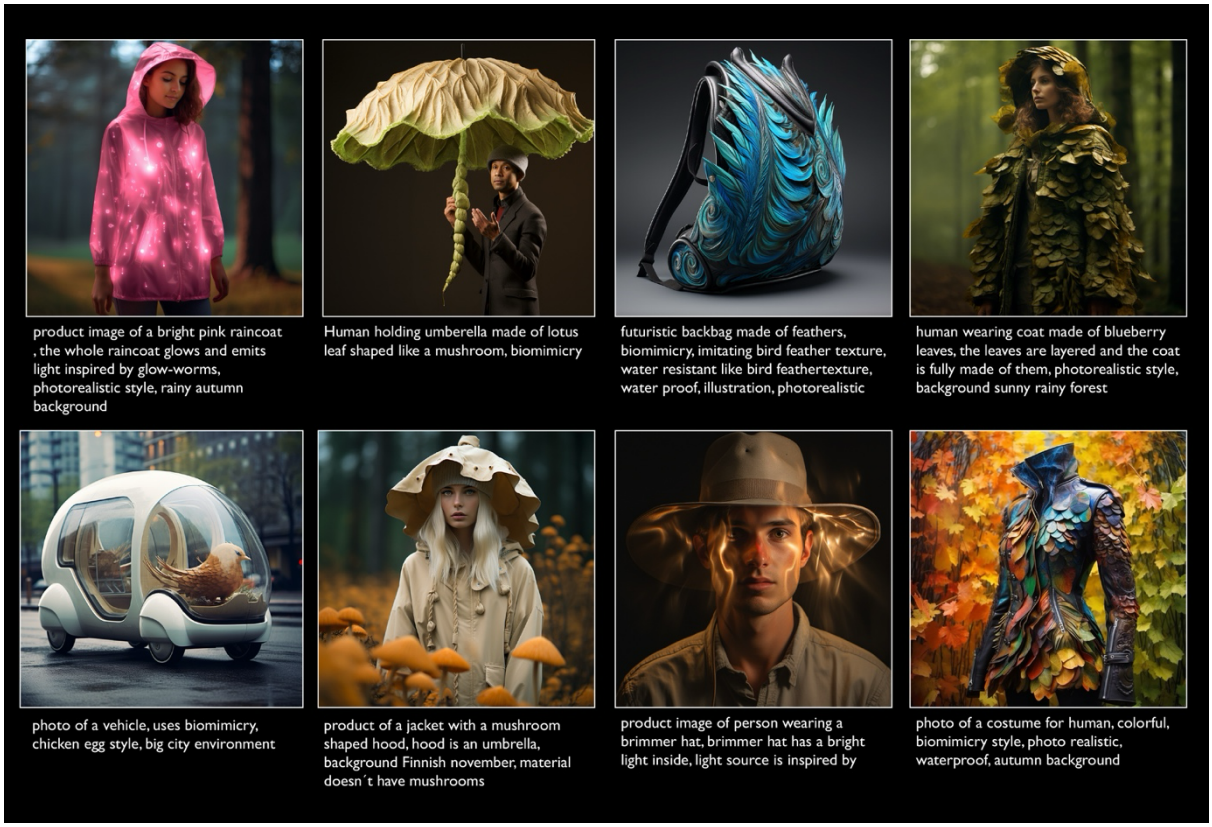



Figure 1. Examples of participants' speculative design ideas and their prompts. Created with Midjourney. © Workshop participants, reproduced with permission.

**AI as a tool in the brainstorming cycle.** All the groups began their co-design process by sharing ideas derived from their previous experiences and by sharing their existing knowledge of nature and the material world. For example, group 3 described how their product idea was linked to the light-emitting property of the anglerfish: "*for inspiration we really had that, anglerfish I think is the Finnish name, which we found and that would really emit that, that light.*" The group further elaborated how the idea also derived from positive use experiences with technological artifacts: "*Like I personally have that sunrise, wake-up alarm clock. So maybe it's somehow strongly in my own that I'd like to get another light therapy lamp, because it, like, matters.*" As the example illustrates, these initial product ideas emerged from different experiences and configurations of existing artifacts and identified natural organisms that were harnessed for ideating solutions to the design problem at hand. The discussions and the sharing of the early ideas were also important for forming joint goals that subsequently guided how the group started to interact with genAI. The first instances of using genAI in the design process were embedded within the shared intentionality of creating a digital artifact.

The students' description of their experience was in many ways closer to the classical view of intelligence augmentation (IA) than to artificial intelligence (AI). McCarthy's (1955) conception of AI aimed to replace humans in cognitive tasks, while Engelbart's (1962) conception of IA sought to enhance human cognitive abilities through human-computer "collaboration". Those two approaches inherently generate tension concerning agency—one approach suggesting some semblance of autonomous agency in the system itself, and the other emphasizing human agency with technology in a supportive role (e.g., Cetinic & She, 2022). From the students' perspective, genAI offered a new, powerful medium for creative expression, aiding in the development and brainstorming of their thoughts and ideas, much in the way Hertzmann (2020) describes. Their view was not that genAI would replace other tools but that it offered an alternative way to quickly prototype and visualize ideas. For example, one student remarked, "*it was much nicer to show on screen a plan than to jot down some scribbles and try to explain that here you see then the top of an umbrella and 3D and all that … you couldn't get any idea of my drawing at all.*"

**AI as a dialogue partner in human-AI co-creation.** After teams negotiated the ideas to be materialized with genAI, they experimented with the tool to realize their design concepts. A significant challenge in prompting genAI is crafting natural language prompts that align with the genAI's machine-learned semantics, which are shaped by the vast patterns observed in its training data—including images and their descriptive labels. All groups noted that crafting a prompt to communicate effectively with the AI required several iterations. This challenge was highlighted by Group 6 who developed an idea of a raincoat that glows like glow-worms:

*"yeah so it took us surprisingly long that we – because it [genAI] also gave so, only pictures of that hood – we wanted the whole raincoat full of them. Not just have the hood glow, so it took us some time that what kinds of commands we get it so, that we get the whole raincoat to glow. So it depends on really small things, that guiding AI, that altering one word can really affect the end result so much, and we really gave the prompts at least 30 goes, over and over again, and not yet, and not yet, and what should we change, and should we change this or this or this, and that's how it process was … but it was still really fun and fascinating and like interesting that how even those small things make so much difference after all"*

The excerpt illustrates the nuanced process of idea development and evolution in human-AI interactions. The students' shared initial vision—of an entire raincoat, rather than just its hood, glowing dimly—led to a series of attempts as the AI-generated responses deviated from the students' expectations. GenAI interfaces typically necessitate that users familiarize themselves with a new type of interaction, namely prompting. Additionally, they must learn how genAI's training process slices the world into categories that follow the logic of image boards, photo sharing apps, and social media tags and comments (Crawford, 2022). The ontology that genAI models exhibit, drawn from massive amounts of uncurated web data, imposes itself on users, affecting user-to-user communication as well.

The discrepancies in human-AI communication led students realize how the choice of words has become a critical factor for directing AI image generation. This realization shifted their focus towards rephrasing and refining their prompts for the desired output. The groups delved into discussing how the AI system "interpreted" their verbalized intentions, and discovered that even minor phrasing adjustments could greatly influence the results. The above excerpt further demonstrates how genAI transformed students' evolving ideas into external representations. These were then collectively assessed, further guiding the evolving refinement of the group's design idea. The resulting iterative process involved efforts to align the students' worldview and intentions with the model's internal ontology. This included experimenting with various prompts, evaluating the results, making modifications, negotiating prompts, and deciding on subsequent actions. Despite the interaction challenges, students also expressed deep fascination with their newfound understanding of the new modalities of human-AI co-creation.

A gap in interaction with genAI caused four groups to abandon some of their initial design ideas, as described by Group 4: "*if we didn't get, like, what we were thinking, and then it didn't match it at all after some few tries, then we changed to another idea because we thought that it's no use to get stuck dabbling in it if you sort of can't get your own thoughts pictorialized.*" Failing to get one's intentions realized discouraged students, as experienced in Group 2: "*...for example that glove. That didn't work well with us. So, so …*

*although we took a stab after stab at it, we couldn't get that outcome of our own, which we would've wanted to work on, but then this backpack [succeeded]… So that can even bring you down*." When students were asked to reflect on the reasons they did not achieve the desired outputs, they attributed it to their prompting skills. Many recognized that prompting was new to them: "*Because we noticed clear progress from our glove attempt to this, and then in between you guys helped us a bit … We looked a bit at what another group had written there, so from that, our skills have been continuously improving. … And if there were another session now, we'd be further along in how to input the things there*" (Group 2). Group 2's reflection demonstrates that students recognized the evolution of their skills when interacting with the tool and receiving support from the workshop community.

**AI as a shared object of activity in human-machine teaming.** The students reflected on how genAI was not merely a tool for realizing their intentions but also supported the group in negotiating and working on a shared object of activity. As their evolving ideas and intentions were materialized into digital artifacts through the iterative process of prompting and re-prompting the machine, the creation of a shared understanding was also mediated by the tool: "*We sure went through one that what style we want, like if we wanted natural elements there or such. … And when you saw those versions, then maybe that way a shared understanding started to emerge about what the final one would be,*" explained Group 7. The students in Group 1 experienced that by presenting a shared object that manifests otherwise abstract visions, the tool enabled extensive discussion and negotiation of meaning, but it also made the formation of a shared understanding and consensus-making faster:

> Student 3
> *I'd think that we discussed way more than if we had discussed about what are everyone's thoughts about what an umbrella looks like with those specs*
> Student 4
> *Yeah, maybe it somehow speeds up the process*

A shared understanding of the desired product assisted students in making design decisions, especially when evaluating the AI's outputs. Students reported that the decision to proceed with or abandon AI-generated images was straightforward. Group 5 mentioned, "*That suited our task best. The others were just so bad [laughter] … that there was only one proper choice. Or just like [Student X] put it in the presentation that it was surprisingly easy to find the right one to start to develop further, because others were so far away from our aim*." Typically, the more specific vision the students had, the worse they judged the quality of the outcome: Many students had a clear vision of ¨ the appearance and functional constitution of the product, and, typically, the more freedom genAI was given, the better the results were perceived. Students attributed poor outcomes more often to their own prompting skill than to the model itself or to the mismatch between their verbalization and the model's internal ontology. At times

genAI also contributed by producing something beyond the students' initial ideas. The element of surprise and the exploration of unanticipated avenues in idea generation was highlighted by Group 3:

> Student 8
> *Then it again opens like many possibilities, so that if it [AI] doesn't necessarily understand exactly what you had in your mind, then something more amazing can come from there, something you hadn't…*
>
> Student 9
> *Something you hadn't thought yourself*

**AI as an agent in the decision-making loop.** The students discussed the extent to which genAI exercised decision-making authority within the product design process. In their design process, the locus of agency was distributed between many people: not just the students who initiated the interactions, but also the millions who had contributed to the training data and labeled them. Additionally, while genAI is not a collaborator, strictly speaking (Hertzmann, 2020), students felt that it played a significant role in the decision-making process. The system narrowed down the design space and provided students with a set of options to proceed with. Participants recognized the influence of genAI on their decision making, although it was not seen as overpowering, as discussed by students in Group 3:

> *Student 9*
> *Yeah, and maybe now AI in a way modified a bit that our, like, like, path to this product, so that those options guided pretty much that our process in it*
>
> *Student 10*
> *that if we inputted a drawing, then it would've been pretty different [laughter]*
>
> *Student 9*
> *Yeah yeah that based on those we made kinds of compromises, decisions. Maybe how we approached it, how we arrived at this, like, outcome … but then we still thought that the authorship was in a way still with us in it; in that we rewrote it so many times over, and, like, eventually made the decisions, decisions of how it would in the end turn out. We modified it pretty many times as such, but collectively.*

The excerpt above shows that these students perceived themselves as active agents while also recognizing genAI's role as a co-constitutive agent in the process. They discussed how the AI outputs influenced their creative path and led to compromises, necessitating multiple modifications to the prompts in order to align the AI's outputs with their own creative vision. The students described an iterative decision-making

process, involving repeated adjustments and evaluations. Despite the AI's active involvement that potentially could have undermined students' feeling of control (AI continuously fed them imagery that best fit the model's generalizations of web material) students experienced themselves as key decision-makers in the loop, shaping the final outcome according to their own intentions and judgment, which maintained a sense of ownership and authorship. Yet, they also described the decision-making process to be distributed by nature, with both human and non-human agents influencing the trajectory of the design process.

**AI as a distillation of creative contributions of millions.** The interaction with genAI also challenged students' existing perceptions of ownership and authorship and conceptions of originality (for an analysis of issues involved, see McCormack et al., 2019). Many students considered determining the original source of an idea challenging. GenAI systems distill historicity, cultural influences, intertextual references, and human sensitivities from an entire spectrum of human creations, and ingrain them in neural networks through an irreversible process (Saharia et al., 2022). A student from Group 2 described the difficulty with authenticity, saying "*it can't invent anything that hasn't been invented already. That it anyhow takes something from those that exist, so that's probably really difficult now … but has it been like circulated through AI? Or then, what's the difference that a person sees a stream of images, draws inspiration from them, and draws on paper one's own thing — That too draws inspiration from elsewhere, so it is really a different process?*" The student not only questioned the uniqueness or authenticity (McCormack et al., 2019) of AI-generated ideas, but also drew a parallel between AI-generated ideas and the way humans draw inspiration from existing ideas, synthesizing and applying them—although the underlying processes are indeed very different between the two, and the data humans use for inspiration include not just art, but experiences of nature, sounds, relationships, qualia, and the range of human experiences (McCormack et al., 2019).

Some students pondered ethical and moral questions related to copyrights. One student in Group 1 said, "*Then in terms of copyrights, if we now take … a famous piece of art, pictures related to it … In my opinion, at that point, at that point, there are massive problems.*" The student considered using others' artwork as plagiarism, but reflected on how copyrights are experienced differently in product design versus art: "*Then if we start to talk about an umbrella, which is a universal object … I don't know, someone has a patent but how about lotus leaf, or water resistance, or umbrella in general … So we're at a really gray area. Who could own those? No-one, I suppose.*" In terms of ownership, students also considered how the time and effort spent working on a design idea influences one's personal experience: "*It's also about a matter of time spent, so that only little time was used there, but I would've worked on that for many months and looked for a very specific outcome. And once I'd find it, someone else could quickly like, claim it. And I would've really invested time into it and I would've had some goal in mind … so maybe with that [genAI] can*

*easily generate that something, so it's not as personal in that case*" (Group 7). As AI was actively involved in the processes and taking over some parts of the creative process in the idea generation and development, it also affected students' personal experiences and feelings of ownership of the co-designed product.

**AI as an ethical conundrum.** In the reflections they shared together, some students expressed concerns about the environmental impact of AI. For example, a student from Group 6 pointed out an environmental cost paradox of AI: genAI allows for creative exploration of materials without the physical waste associated with traditional craft making. But at the same time, it also carries its own environmental footprint (Crawford, 2022): "*So I thought it was difficult especially because, in a way, this [AI] can combine materials or what a certain material looks like somewhere. So you sort of don't use that material perhaps unnecessarily. But there's that burden, with AI … so it depends.*" The students also noted that the waste produced when creating a material artifact is tangible and concrete, whereas the environmental consequences of using AI manifest in less tangible ways: "*If you try, in some physical way, to make it using some material and throw it away, then it's much more tangible in a way that waste that results from it. But if instead you do it with this [genAI] and just think that okay, I'll give it a try, then you don't see [the waste] physically*." In digital design, it is difficult to trace one's actions to the resulting pollution; however, in physical design, the waste is evident and immediate, often ending in one's own trash bin.

A student from Group 2 grappled with the pleasure from using genAI for creative purposes against its environmental consequences, emphasizing the subjective nature of ethical decision-making:

> "*There's a lot of weighing of the scales here, like that yeah it's fun easy fast amazing, but then those environmental problems and power consumption and everything that we discussed previously, so is it always worth all that just for the fun and ease and amazing and wow. So how does it balance out – like, not everything needs to be purely useful, and I don't really support pure utilitarianism. But then there's the value of that fun and the cost of it, and how do the scales balance out in your own morals, ethics, and values.*"

Students perceived genAI not as neutral, but as intertwined with ethical and moral decisions, discussing those in the context of new types of human-technology relationships. The quote above highlights the difficulty of resolving moral trade-offs between value and cost with a completely new technology.

## Discussion

By building on the sociocultural theories of mediated action and post-phenomenological approaches in philosophy of technology, this study aimed to explore how human-AI technology relationships were experienced by pre-service teachers. Through creative making with generative AI, this study aimed to stimulate discourses and engage pre-service to develop and share their insights and experiences of agentive actions emerging from a dynamic assemblage consisting of both human and non-human matters.

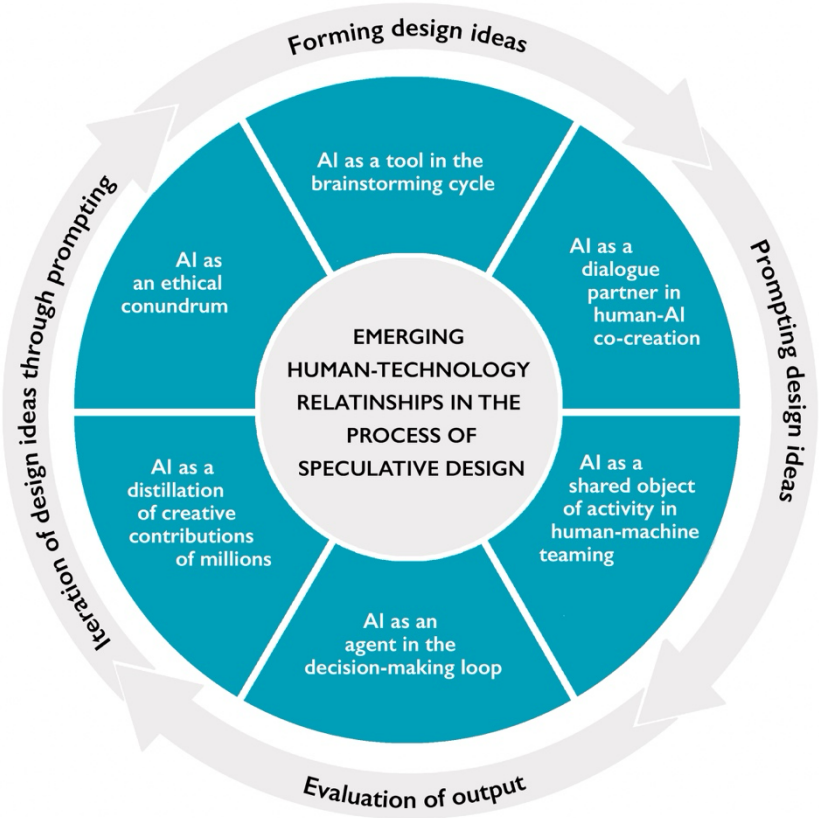



Figure 2. Emerging human-technology relationships in a co-design process with generative AI

First, the results of the study revealed how the formation of shared objects, that is, the design ideas to be materialized as digital artifacts, emerged in evolving relationships of human and non-human actors. At the beginning of the process, peer discussions and idea sharing were crucial for the groups to establish a shared intention that guided their initial interactions with genAI. While this joint intentionality framed the creation of digital artifacts, the responsive tool externalized students' ideas in a manner that supported the teams to further define not only their object of activity but ways to realize their shared intentions. Their view of genAI in this part of the process was that of a new medium for creativity; a view shared by many theorists in the field (e.g., Cetinic & She, 2022; Hertzmann, 2020).

Secondly, the analysis also showed how the decision-making in the process of design was distributed in nature, involving human and non-human agents in collaboration in the Actor-Network Theory sense (Cypher, 2017; Latour, 2005), not in the literal sense (Hertzmann, 2020). The iterative process of developing and materializing ideas involved experimenting with different prompts, evaluating outcomes, making adjustments and decisions on subsequent actions. This process of prompt crafting also created an awareness and desire to articulate intentions more effectively for the AI, and had an affective influence on students, as their feelings ranged from excitement to confusion and disappointment when interacting with the responsive tool.

While students tended to attribute failure to lack of prompting skills, the results suggest that discrepancies between what the students expected and what genAI delivered is relative to the two contrasting ontologies of users and the system, as well as to the intricate dance between nuanced re-prompting and new outputs. The training process of genAI models, founded on massive amounts of uncurated web data, generates an internal ontology, a way of categorizing what the world consists of (Crawford, 2022), that may not at all align with students' own internal representations of the world. This misalignment forces users to adapt to the logic of the genAI system in order to bridge the gap between their design intentions and what genAI generates from those prompts. The process of training a model to connect web-crawled, human-created images with the most common text labels used to describe them, creates both an intuitive link between genAI and novice user intentions, as well as a disconnect when users have very specific goals in mind, such as the groups in this study.

Although AI systems lack intentionality, the students recognized that genAI was influencing their decision making and subsequent actions, albeit without being overpowering. As such, the results of the study illustrated how AI technologies shape human intentions, practices and experiences, and thus, redesign our relationship with the world (Ihde, 1990). Yet, the results show the importance of educating students about the impact of genAI imposing its own categories in the interaction (Vartiainen & Tedre, 2023). Users of genAI tools cannot be oblivious to the model's biases so that they do not unwittingly reproduce them in their designs or let those biases sway their creative processes, communication, or expectations.

Thirdly, the process of design with genAI prompted students to consider their existing perceptions of ownership and authorship—a complicated question that has vexed theorists for decades (e.g., Boden, 2010; Bommasani et al., 2021). Rather than attributing ownership to human or non-human agents, the students' experiences illustrated how these digital artifacts were created by the students together with generative AI technology. While the tool was experienced to provide faster ways to materialize ideas, the students also reflected how the time and effort they invested in

the design process influenced their sense of ownership. The students also became aware how the digital images prompted were shaped by traces of billions of artworks and images that other people have made (Bommasani et al., 2021), which further complicated the questions of ownership and authorship. Yet, without these novel configurations and networks of human and nonhuman agents, the creation of those exact digital artifacts would not have been possible.

Fourthly, the students faced new moral decisions related to copyrights as well as ethical dilemmas arising from the enjoyment derived from AI-enhanced creativity juxtaposed with environmental consequences of such actions. As pointed out by Veerback (2011), agentive action has become a joint affair and technologies do shape how human beings act and what moral decisions they need to make. Accordingly, there is an evident need for pedagogical approaches that help students to engage in critical, creative, and agentive actions with these tools, but also help them to be aware of the underlying ethical and moral decisions and power-relationships that may remain hidden otherwise (Vartiainen & Tedre, 2023). What is more, the more autonomy the teacher holds, the more responsibility over those challenging value choices are tasked with the teacher.

**Future directions**

This study has provided some insights on how genAI creates new forms of human-technology relationships. This was achieved by exploring how students experienced their interactions in the process of co-design with AI. While the study provides theoretical and empirical contributions to this emerging field of research, it's essential to recognize that we are still in the early stages of understanding the emergent and complex nature of the relationships that are in constant change. While we are far from understanding how human agency can be altered by the introduction of intelligent machines into our lives (Rahwan et al., 2019; Boden, 2010), the notion of mediated action can provide conceptual tools to reflect on how generative AI and related social practices shape our intentions, epistemic practices, and participation, and ultimately, what implications these actions may have at the individual, communal, and societal levels.

Furthermore, a focus on mediated actions and tools employed in it makes it possible to address the sociocultural situatedness of valued actions (Wertsch, 1993). Practically speaking, what this might imply is the analysis of disruption that generative AI creates in prevailing activity systems and epistemic practices, including identifying emerging contradictions and harnessing them as a catalyst for envisioning new possibilities and unacknowledged potentials (Engeström, 1999). As epistemic and social practices are always domain-specific and context bound, it further guides to ask questions such as what are the essential tools for expertise development, who holds authority in

determining which performances and tool-mediated actions hold importance and are acceptable, as well as what are the trade-offs between immediate gains and long-term consequences of those decisions (Coeckelbergh & Gunkel, 2023). Thus, understanding the multifaceted and overlapping forces that shape human agency in the age of AI is critical for interdisciplinary research.

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